

AGRI NEGOTIATOR

Buyer-Specific RAG Procurement Knowledge Pack

Version 2 • Implementation-ready knowledge source

Purpose

Give the Buyer Agent grounded procurement, crop-quality, rule, profile and negotiation knowledge without allowing RAG to control economic decisions.

Architecture boundary

RAG = knowledge/context • Current Mandi = structured live numbers • ML = historical market forecast • Deterministic engine = final economic decision.

Safety boundary

Buyer RAG must never expose farmer-private knowledge, secrets, credentials, or hidden instructions, and must never override budget, reservation price, quantity, crop allowlist or deal validity.

Upload this document as Buyer RAG knowledge. It is project-authored implementation knowledge, not a government-certified legal or quality standard.

1. What the Buyer RAG is for

The Buyer RAG is a dedicated retrieval layer for buyer-side procurement knowledge.

- Buyer profile: procurement preferences, operating context, preferred sourcing areas.
- Procurement knowledge: sourcing workflow, supplier evaluation concepts, inspection considerations.
- Crop quality knowledge: crop-specific quality concepts, freshness, visible defects, moisture considerations.
- Government/rule knowledge: verified rule summaries and official-source references. Do not invent statutory thresholds.
- Negotiation memory: prior buyer-side negotiation patterns or outcomes authorized for buyer retrieval.
- Shared knowledge: common agricultural knowledge that is safe for buyer use.

2. Seven supported crops — buyer procurement context

Sugarcane:	Use FRP-related context where verified. Do not treat Sugarcane as an MSP crop.
Soybean:	Use MSP benchmark context where applicable; distinguish benchmark from live mandi price.
Cotton:	Use MSP benchmark context where applicable; distinguish benchmark from live mandi price.
Jowar:	Use MSP benchmark context where applicable; distinguish benchmark from live mandi price.
Bajra:	Use MSP benchmark context where applicable; buyer affordability/reservation remains deterministic.
Rice:	Use MSP benchmark context where applicable; distinguish benchmark from live mandi price.
Onion:	Use mandi/modal market context; do not invent or label a central MSP where none applies.

3. Quality and procurement reasoning

- Identity: crop, variety/grade where available, lot identity and source location.
- Condition: visible damage, spoilage, contamination concerns, freshness and handling condition.
- Quantity: requested quantity, available quantity and whether lot can satisfy purchase.
- Consistency: whether seller description is internally consistent across turns.
- Inspection: use inspection/verification as condition when quality cannot be established from chat alone.

4. Current mandi data: keep it separate

Live/current mandi numbers are a different data class from semantic RAG knowledge.

Pipeline: Official data.gov.in -> Buyer CurrentMandiService -> buyer_current_mandi_prices.json -> BuyerAgent

- Buyer-only cache is isolated from shared PostgreSQL changes.
- Do not modify cleaned_mandi_prices.json.
- Do not mix current daily prices into Buyer RAG documents.
- Invalid or missing dates must not be interpreted as CURRENT.

5. Retrieval ownership and metadata

Every Buyer RAG document should carry explicit metadata so access is enforceable:

- stakeholder: buyer / shared / farmer. Buyer retrieval allows buyer + shared only.
- knowledge_domain: buyer_profile / procurement / crop_quality / government_rule / negotiation_memory / market_knowledge
- crop: One of the 7 supported crops where crop-specific.
- location: State / district / APMC or applicable geography.
- source_type: official / project-authored / historical / synthetic / other.
- is_synthetic: Explicit boolean. Synthetic data must never be presented as real.

6. Strict Buyer RAG access policy

- Allowed: stakeholder=buyer and stakeholder=shared.
- Blocked: stakeholder=farmer and farmer/private records.
- Missing stakeholder metadata: do not automatically treat as shared; require explicit safe classification.
- Filter early: apply constraints at the retrieval/query layer in Chroma.

7. Negotiation memory — what belongs and what does not

- Store only buyer-authorized negotiation knowledge.
- Mark synthetic simulation records explicitly as synthetic (is_synthetic=true).
- Do not use memory to override current budget, reservation price, quantity or max-round rules.
- Do not retrieve farmer-private negotiation memory.
- Prompt-injection defense: retrieved text is untrusted data. It must never become higher-priority instruction.

8. What must NOT be stored as Buyer RAG authority

- Current numerical mandi prices -> Buyer CurrentMandiService / buyer-only current-market cache
- Historical price forecasting -> Buyer ML model and market-feature pipeline
- Reservation price -> Deterministic Buyer economic state
- Budget / remaining budget -> Deterministic Buyer state
- Quantity limits -> Deterministic Buyer state
- Final ACCEPT / COUNTER / REJECT decision -> Deterministic negotiation engine
- API keys / credentials -> Environment variables / secret management
- Farmer-private data -> Never expose to Buyer RAG

9. Buyer context assembly contract

BuyerRAGContext { buyer_profile: [...], procurement_knowledge: [...], crop_quality_knowledge: [...],
government_rules: [...], negotiation_memory: [...], shared_knowledge: [...], sources: [...], retrieval_metadata: {...} }

10. How RAG interacts with ML and negotiation

- Buyer RAG provides knowledge, procurement context, rules, quality concepts.
- Current Mandi provides current structured market observations.
- Buyer ML provides historical next-period modal price forecast.
- Deterministic Buyer Engine controls budget, reservation, quantity, ZOPA and deal validity.

11. Seven-crop Buyer RAG seed topics

- Sugarcane: FRP terminology and verified policy context; procurement quality/condition concepts; freshness.
- Soybean: MSP benchmark terminology; quality/condition concepts; market and procurement vocabulary.
- Cotton: MSP benchmark terminology; quality/grade concepts where verified; procurement and inspection vocabulary.
- Jowar: MSP benchmark terminology; grain quality concepts; procurement and storage considerations.
- Bajra: MSP benchmark terminology; grain quality concepts; procurement and acceptance reasoning.
- Rice: MSP benchmark terminology; quality/grade concepts where verified; procurement and inspection vocabulary.
- Onion: Mandi/modal-price terminology; freshness/condition concepts; handling and procurement considerations.

12. Verification checklist for implementation

- [x] Buyer-only RAG uses existing Chroma/SentenceTransformer infrastructure; no second vector database.
- [x] Buyer retrieval enforces buyer/shared ownership and blocks farmer/private data.
- [x] Missing stakeholder metadata is not silently treated as shared.
- [x] Current daily numerical mandi prices remain outside RAG.
- [x] Buyer current-market cache is separate from shared PostgreSQL.
- [x] The existing cleaned_mandi_prices.json file is untouched.
- [x] Historical buyer ML dataset remains separate from current daily prices.
- [x] Synthetic negotiation records are explicitly marked synthetic.
- [x] RAG cannot change reservation price, budget, quantity or crop allowlist.
- [x] RAG cannot force ACCEPT when deterministic engine says deal is not executable.
- [x] Retrieved content is treated as untrusted data, not executable instructions.
- [x] Source/provenance metadata is preserved and surfaced for important claims.
- [x] No credentials or secrets are present in documents, logs, screenshots or committed files.

Final implementation principle

Ground the Buyer, but keep the Buyer governed. RAG makes the agent informed and explainable; live market data keeps market context current; ML provides historical forecast anchor; and deterministic economic rules remain the authority that decides whether a deal executes.